SALES PERFORMANCE ANALYSIS

SQL Data Analysis & Power BI Dashboard Project

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Project Type: **Data Analytics Portfolio Project**

Tools: **MySQL, SQL, Microsoft Power BI, DAX, Microsoft Word**

Dataset: **Sample Superstore**

Analysis Reference Date: **October 18, 2026**

PROJECT SNAPSHOT

Key Performance Indicator	Result
Total Sales	2,326,534.35
Total Profit	292,296.82
Overall Profit Margin	12.56%
Distinct Orders	5,111
Total Quantity Sold	Approximately 39,000 units
Sales Records Analyzed	10,194
Records Flagged as Future-Dated	1,083
Future-Dated Records as a Share of Dataset	10.63%

Note: Future-dated records are identified relative to the fixed analysis reference date of October 18, 2026. This classification does not automatically mean that the records are incorrect.

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1. Executive Summary

This project analyzes retail sales performance using SQL and Microsoft Power BI. The objective is to evaluate sales, profitability, regional performance, product performance, discount patterns, and selected data-quality indicators to identify business opportunities and areas requiring further investigation.

The dataset contains 10,194 sales records, alongside customer-region information and product-return information. The analysis combines SQL-based investigation with an interactive Power BI dashboard to transform transactional data into meaningful business insights.

The dataset generated total sales of **2,326,534.35** and total profit of **292,296.82**, resulting in an overall profit margin of approximately **12.56%**. The West region generated the highest sales and profit among the four regions analyzed. Technology achieved the highest category-level profit, while Furniture generated relatively low profit compared with its sales volume.

The sub-category analysis identified several areas requiring attention. Tables, Bookcases, and Supplies recorded negative overall profits. The discount analysis also revealed a substantial profitability difference between transactions with no discount and transactions with high discounts. The high-discount group generated a total loss of 136,020.69.

A data-quality analysis identified 1,083 records with order dates later than the fixed reference date of October 18, 2026. These records represent approximately 10.63% of the sales dataset and should be reviewed to determine whether they are valid future transactions, planning records, or date-quality issues.

The final Power BI dashboard presents three interactive pages: Sales Overview, Product and Profitability Analysis, and Discount and Data Quality Analysis. Together, they provide an accessible view of business performance and support evidence-based recommendations.

2. Project Objectives

The main objectives of this project are:

- Analyze total sales, profit, order volume, and quantity sold.
- Compare sales and profitability across geographic regions.

- Evaluate the performance of product categories and sub-categories.
- Investigate the relationship between discounts and profitability.
- Identify products and sub-categories associated with negative profit.
- Examine selected transaction-level patterns.
- Flag records dated after a fixed analysis reference date.
- Build an interactive Power BI dashboard using DAX measures and calculated columns.
- Translate analytical findings into practical business recommendations.

3. Dataset Overview

3.1 Data Sources

The project uses three tables associated with the Sample Superstore dataset.

Table	Number of Rows	Purpose
Orders	10,194	Sales transactions, dates, products, quantities, discounts, sales, and profit
People	4	Regional information
Returns	296	Returned-order information
Total rows across tables	**10,494**	**Combined source tables**

The number of rows across all three tables is not the same as the number of sales records because each table serves a different purpose.

3.2 Key Fields

The Orders table contains fields such as:

- Order ID
- Order Date
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

The People table contains regional information, while the Returns table identifies orders recorded as returned.

3.3 Date Coverage

The available order dates range from January 3, 2023, to December 30, 2026.

The data includes transactions dated after October 18, 2026. These records are retained for analysis and separately flagged using a fixed reference date so that they can be investigated.

4. Methodology

The analysis followed a structured data analytics workflow.

4.1 Data Inspection

The source tables were inspected to understand their structure, row counts, fields, and relationships. Relevant columns were reviewed for sales, profit, discount, product, regional, and date analysis.

4.2 SQL Analysis

MySQL was used to investigate the dataset and calculate business metrics. SQL operations included:

- SELECT and WHERE for retrieving relevant records.
- GROUP BY for summarizing results by region, category, sub-category, and discount level.
- Aggregate functions such as SUM, AVG, and COUNT.
- CASE expressions for categorizing discounts.
- Joins for combining related tables.
- Conditional filters for identifying loss-making products and selected transactions.

4.3 Power BI Data Modeling

The data was imported into Power BI, and relationships were established between the regional information, orders, and returns tables.

The model was used to support interactive filtering and consistent calculations across dashboard pages.

4.4 DAX Calculations

DAX measures were created to calculate sales, profit, profit margin, distinct orders, quantity, and the number of future-dated records. Calculated columns were added to classify discount levels and date status.

4.5 Dashboard Development

The dashboard was organized into three pages:

1. Sales Overview

2. Product and Profitability Analysis

3. Discount and Data Quality Analysis

Each page focuses on a specific analytical objective while maintaining consistent formatting, colors, and layout.

4.6 Business Interpretation

The results were interpreted to identify areas of strong performance, possible profitability risks, and opportunities for further investigation. Recommendations are based on observed associations and aggregate results rather than assumptions about causation.

5. Sales Performance by Region

Regional analysis helps compare the geographic distribution of sales and profit.

Region	Total Sales	Total Profit	Profit Margin
West	739,813.61	110,798.82	14.98%
East	691,828.17	94,883.26	13.71%
Central	503,170.67	39,865.31	7.92%
South	391,721.91	46,749.43	11.93%

5.1 Key Findings

The West region recorded the highest sales and total profit. Its profit margin of 14.98% was also the highest among the four regions.

The East region ranked second in sales and profit, with a margin of 13.71%.

The Central region generated 503,170.67 in sales but only 39,865.31 in profit. Its margin of 7.92% was the lowest regional margin in the analysis.

The South region recorded the lowest sales but generated more profit than the Central region. This illustrates why sales volume alone does not fully explain business performance.

5.2 Business Interpretation

The West and East regions provide useful reference points for evaluating regional performance. The Central region deserves additional investigation to understand its relatively low profitability.

Further analysis could examine differences in product mix, discount levels, shipping costs, and other expenses. These factors should be tested before attributing the margin differences to any single cause.

6. Profit Margin by Region

Profit margin measures the share of sales revenue retained as profit.

Formula:

Profit Margin = Total Profit ÷ Total Sales × 100

The regional margins are:

- West: 14.98%
- East: 13.71%
- South: 11.93%
- Central: 7.92%

The West region retains approximately 14.98 in profit for every 100 in sales, based on the reported totals. The Central region retains approximately 7.92 per 100 in sales.

The difference indicates that regional performance should be evaluated using both revenue and profitability metrics. A region can generate substantial sales without achieving a comparable profit margin.

Recommendation: Review the Central region's product mix, discount distribution, and order-level profitability. Compare the findings with the West and East regions to identify possible improvements.

7. Sales and Profitability by Category

The category analysis evaluates the financial contribution of Technology, Furniture, and Office Supplies.

Category	Total Sales	Total Profit	Profit Margin
Technology	839,893.28	146,543.38	17.45%
Furniture	754,747.76	19,730.00	2.61%
Office Supplies	731,893.31	126,023.44	17.22%

7.1 Technology

Technology generated the highest sales and total profit. Its profit margin of 17.45% was slightly higher than that of Office Supplies.

This category represents a significant contributor to the dataset's overall profitability.

7.2 Office Supplies

Office Supplies generated 731,893.31 in sales and 126,023.44 in profit. Its margin of 17.22% indicates a strong profit contribution relative to its sales.

7.3 Furniture

Furniture generated 754,747.76 in sales but only 19,730.00 in profit, resulting in a margin of 2.61%.

Although its sales were substantial, the category's profitability was comparatively low. The sub-category analysis provides additional detail on this result.

7.4 Key Finding

Technology and Office Supplies generated considerably higher profit margins than Furniture. Furniture's low margin suggests that its product mix, discounting practices, and costs warrant closer investigation.

8. Sales and Profitability by Sub-Category

Sub-category analysis provides a more detailed view of product performance and helps identify areas that may be hidden by category-level totals.

8.1 Selected Loss-Making Sub-Categories

Sub-Category	Sales	Total Profit	Profit Margin
Tables	208,020.18	-17,753.21	-8.53%
Bookcases	115,361.20	-3,632.07	-3.15%
Supplies	46,725.50	-1,171.39	-2.51%

Tables generated the largest overall loss among these three sub-categories. Bookcases and Supplies also recorded negative profit.

Negative profit margins indicate that the aggregated profit for the selected sub-category was below zero over the analyzed transactions.

8.2 Selected Profitable Sub-Categories

Sub-Category	Total Profit	Profit Margin
Copiers	56,093.94	37.21%
Binders	31,426.10	15.16%

Copiers recorded a strong profit contribution and a high margin. Binders also generated positive profit, despite having a relatively high average discount in the analyzed data.

8.3 Key Findings

The analysis shows that profitability varies substantially between sub-categories. High sales volume does not guarantee positive profit, as illustrated by Tables.

Recommendation: Review the pricing, discounting, and cost structure of loss-making sub-categories. Decisions should be based on product-level and order-level evidence rather than sub-category averages alone.

9. Discount and Profitability Analysis

Discounts can help stimulate demand, but excessive discounting may reduce profit. This analysis investigates how discount levels are associated with profitability.

The discount classification used in Power BI is:

Discount Level	Rule
No Discount	Discount equals 0
Low	Discount greater than 0 and up to 10%
Medium	Discount greater than 10% and up to 20%
High	Discount greater than 20%

9.1 Interpretation

The categories provide a consistent way to compare transactions with different discount levels.

The analysis shows that profitability differs substantially across these groups. Transactions in the High discount group recorded a negative total profit, while the No Discount group generated the largest positive total profit.

However, these groups may contain different products, order sizes, and customer purchasing patterns. The results demonstrate an association between discount levels and profitability; they do not independently prove that discounts are the sole cause of losses.

10. Identification of Loss-Making Products

The product-level analysis identifies individual products with negative aggregated profit. The following examples illustrate the types of records requiring closer investigation.

Product	Number of Rows	Average Discount	Sales	Total Profit
Cubify CubeX 3D Printer Double Head Print	3	53.33%	11,099.96	-8,879.97
Lexmark MX611dhe Monochrome Laser Printer	4	40.00%	16,829.90	-4,589.97
Cubify CubeX 3D Printer Triple Head Print	1	50.00%	7,999.98	-3,839.99
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	5	28.00%	9,917.64	-2,876.12

These examples show that some products generated substantial sales while recording negative profit. The high average discounts on several of these products make discount policy one relevant area for investigation.

The examples above are selected records from the analysis, not an exhaustive list of every loss-making product.

10.1 Business Implications

The business should review:

- Whether discounts were necessary to generate each sale.
- Whether the selling price adequately covered product and fulfillment costs.
- Whether particular products consistently generate losses.
- Whether loss-making transactions reflect special circumstances or a recurring pattern.

Any change to product pricing or discount policy should be evaluated against customer demand and overall profitability.

11. Overall Profitability by Discount Level

This section compares the four discount groups using transaction counts, average discounts, sales, profit, and profit margin.

Discount Level	Rows	Average Discount	Average Profit per Row	Total Sales	Total Profit	Profit Margin
No Discount	4,925	0.00%	66.34	1,105,323.79	326,718.59	29.56%
Low	96	10.00%	94.79	54,952.50	9,099.97	16.56%
Medium	3,758	19.93%	24.61	801,497.92	92,498.94	11.54%
High	1,415	58.33%	-96.13	364,760.15	-136,020.69	-37.29%

11.1 No Discount

The No Discount group contained 4,925 rows and generated total profit of 326,718.59. Its margin of 29.56% was the highest of the four groups.

This group represents a large share of the dataset and provides a useful benchmark for comparison.

11.2 Low Discount

The Low discount group contained 96 rows. Its total profit was 9,099.97, with a margin of 16.56%.

Because this group is much smaller than the other groups, its average results should be interpreted with caution.

11.3 Medium Discount

The Medium discount group contained 3,758 rows and generated 92,498.94 in profit. Its margin of 11.54% was lower than the margins of the No Discount and Low groups.

11.4 High Discount

The High discount group contained 1,415 rows and generated a total loss of 136,020.69. Its profit margin was -37.29%.

This is the most significant negative profitability result in the discount analysis. It suggests that transactions receiving high discounts should be examined more closely.

11.5 Key Finding

The High discount group recorded a substantial loss, while the No Discount group generated the largest total profit.

These results support reviewing high-discount transactions, particularly when they involve products or sub-categories already showing negative profitability. Further investigation is needed to determine the contribution of discounting, product costs, and other factors.

12. Transaction-Level Investigation

Aggregate metrics reveal overall patterns, but transaction-level investigation helps identify the records behind those patterns.

The analysis can be extended by filtering orders based on:

- Product name and sub-category.
- Discount percentage.
- Sales and profit values.
- Region and order date.
- Returns information, where applicable.

For example, a high-discount transaction with negative profit can be investigated alongside other transactions for the same product. This can help determine whether the loss is isolated or recurring.

A transaction-level review should distinguish between a negative-profit transaction and a product that is consistently unprofitable across multiple transactions.

The Power BI dashboard supports this investigation by allowing users to filter and compare sales, profitability, and discount-related results.

13. Data Quality Assessment

Data-quality checks are important because incorrect or unexpected dates can affect reporting, trend analysis, and business decisions.

13.1 Date Status Classification

A calculated column named Date Status was created in Power BI to classify order dates relative to October 18, 2026.

Records with an order date later than the reference date are classified as Future-Dated. All other records are classified as Valid Date for the purposes of this specific check.

The label Valid Date means only that the order date does not exceed the chosen reference date. It does not confirm that every other field in the record is valid.

13.2 Results

Indicator	Result
Total Sales Rows	10,194
Future-Dated Rows	1,083
Share of Rows Flagged	10.63%
Sales in Future-Dated Rows	238,474.85
Profit in Future-Dated Rows	22,886.14

The 1,083 flagged records are distributed across October, November, and December 2026.

Order Month	Rows	Sales	Profit
October 2026	141	34,845.31	4,538.21
November 2026	460	118,454.51	9,692.10
December 2026	482	85,175.03	8,655.83

Total	**1,083**	**238,474.85**	**22,886.14**

13.3 Interpretation

The flagged records contribute both sales and profit to the dataset. Their presence should not automatically be treated as evidence of incorrect data.

Depending on the intended purpose of the dataset, these dates may represent valid future transactions, forecast-like records, or date issues. Their meaning should be confirmed before excluding them from business reporting.

13.4 Limitations of the Check

The date classification uses a fixed reference date of October 18, 2026. If the reference date changes, the number of records classified as future-dated may also change.

The classification does not automatically detect missing values, duplicated business transactions, inconsistent product information, or incorrect financial values. Those issues require separate validation rules.

Recommendation: Confirm the intended reporting period and dataset semantics before deciding whether to retain, exclude, or separately report the flagged records.

14. Power BI Dashboard

The final Power BI dashboard consists of three pages designed to present the analysis in a clear, interactive format.

14.1 Sales Overview

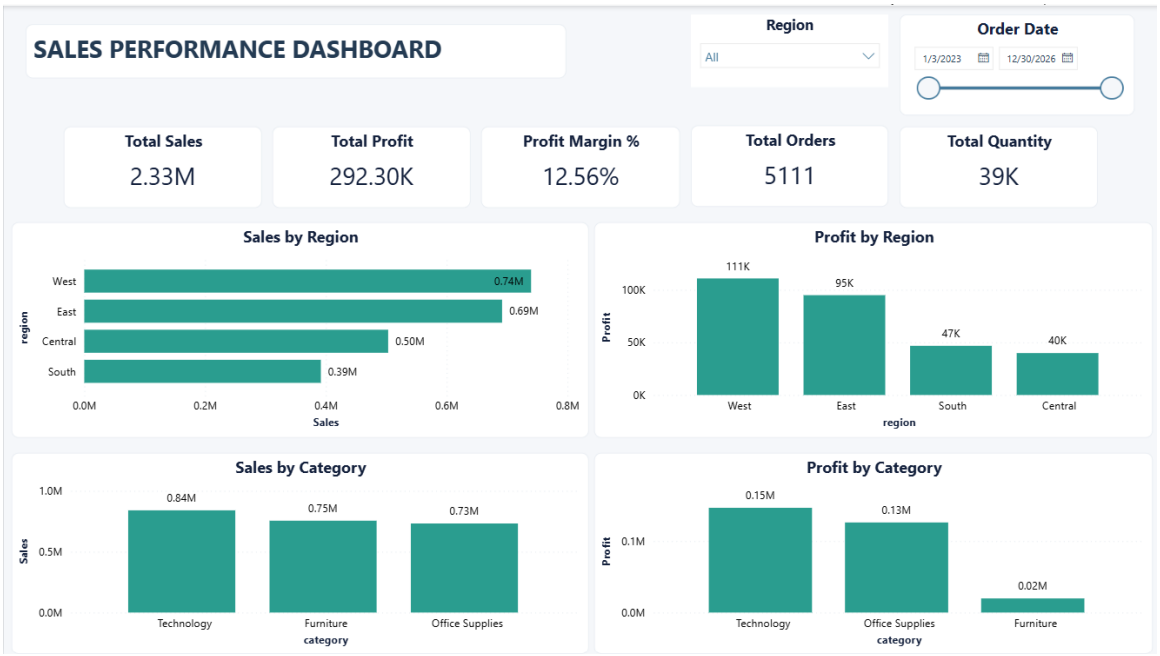


Figure 1. Sales Overview Dashboard

The Sales Overview page presents five key performance indicators:

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Total Quantity

It also includes regional and category-level charts, along with slicers for Region and Order Date.

The page provides a high-level summary of business performance and allows users to investigate how the main KPIs change across selected regions and time periods.

Key insight: The West region generated the highest sales and profit, while Technology generated the highest category-level profit.

Dashboard screenshot: images/01_Sales_Overview.png

14.2 Product and Profitability Analysis

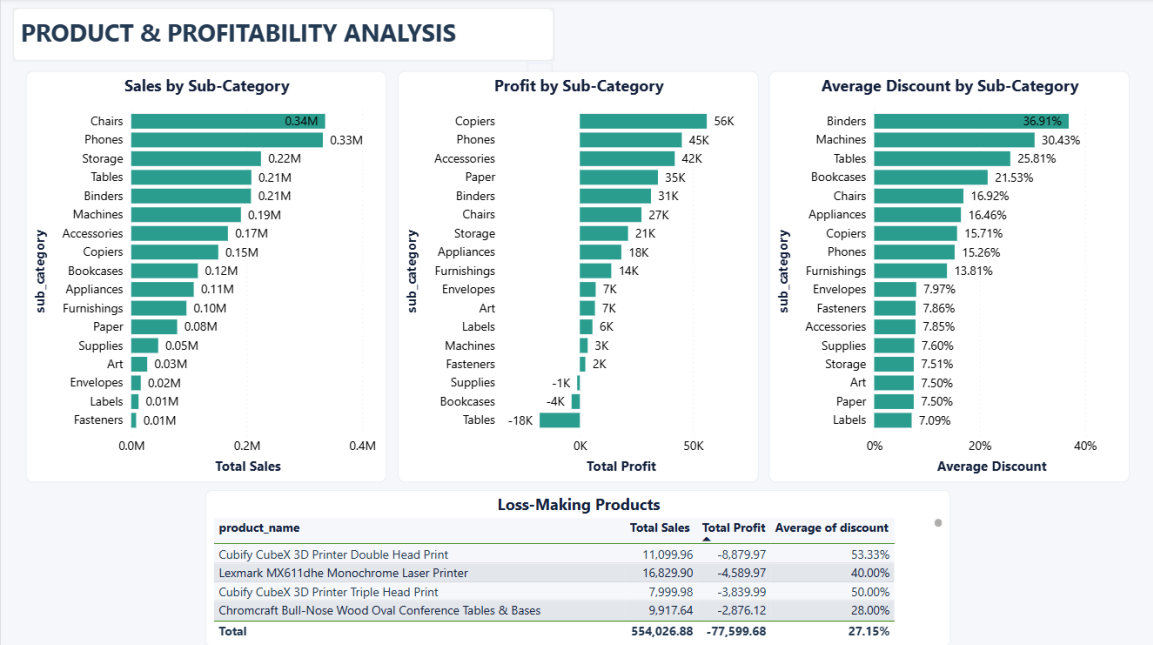


Figure 2. Product and Profitability Analysis Dashboard

The second page focuses on sub-category performance and loss-making products.

It includes:

- Sales by Sub-Category.
- Profit by Sub-Category.
- Average Discount by Sub-Category.
- A table of selected loss-making products.

This layout allows the reader to compare sales volume, profit contribution, and discount patterns. The product table provides a more detailed view of selected products with negative profit.

Key insight: Tables recorded a total loss of 17,753.21, while Copiers generated a profit of 56,093.94.

Dashboard screenshot: images/02_Product_Profitability.png

14.3 Discount and Data Quality Analysis



Figure 3. Discount and Data Quality Analysis Dashboard

The third page combines discount analysis with the date-quality check.

It includes:

- Profit by Discount Level.
- A Future-Dated Rows KPI.
- Future-Dated Orders by Month.

The page makes it possible to compare profitability across discount groups while reviewing the distribution of records dated after the fixed analysis reference date.

Key insight: The High discount group generated a loss of 136,020.69. Separately, 1,083 records were flagged as future-dated relative to October 18, 2026.

Dashboard screenshot: images/03_Discount_Data_Quality.png

15. Key DAX Measures and Calculated Columns

DAX was used in Power BI to create reusable measures and calculated columns. Measures calculate results in the current filter context, while calculated columns classify individual rows.

15.1 Total Sales

Total Sales =

SUM('sales_performance orders'[sales])

This measure calculates the sum of sales values. It updates when the user applies dashboard filters.

15.2 Total Profit

Total Profit =

SUM('sales_performance orders'[profit])

This measure calculates the sum of profit values for the selected data.

15.3 Profit Margin

Profit Margin % =

DIVIDE([Total Profit], [Total Sales], 0)

This measure calculates total profit divided by total sales. The DIVIDE function returns zero if the denominator is zero or blank.

The measure should be formatted as a percentage in Power BI.

15.4 Total Orders

Total Orders =

DISTINCTCOUNT('sales_performance orders'[order_id])

This measure counts distinct order IDs rather than counting every sales row. An order may contain multiple product rows, so distinct counting provides a more appropriate order-volume KPI.

15.5 Total Quantity

Total Quantity =

SUM('sales_performance orders'[quantity])

This measure calculates the total quantity across the selected sales records.

15.6 Discount Level — Calculated Column

```
Discount Level =

SWITCH(

    TRUE(),

    'sales_performance orders'[discount] = 0, "No Discount",

    'sales_performance orders'[discount] <= 0.10, "Low",

    'sales_performance orders'[discount] <= 0.20, "Medium",

    "High"

)
```

This calculated column assigns a discount category to each row:

- No Discount: discount equals zero.
- Low: discount is greater than zero and up to 10%.
- Medium: discount is greater than 10% and up to 20%.
- High: discount is greater than 20%.

The resulting classification can be used in charts, tables, and filters.

15.7 Date Status — Calculated Column

```
Date Status =

IF(

    'sales_performance orders'[order_date] > DATE(2026, 10, 18),

    "Future-Dated",
```

```
    "Valid Date"
)
```

This calculated column classifies each row according to whether its order date falls after October 18, 2026.

The reference date is fixed in the formula. It should therefore be documented as an analysis reference date rather than treated as a dynamically changing definition of the future.

15.8 Future-Dated Rows — Measure

```
Future-Dated Rows =
CALCULATE(
    COUNTROWS('sales_performance orders'),
    'sales_performance orders'[Date Status] = "Future-Dated"
)
```

This measure counts rows classified as Future-Dated. It supports the Future-Dated Rows KPI and the related data-quality analysis.

15.9 DAX Implementation Notes

The measures are designed to respond to relevant Power BI filters. The Discount Level and Date Status columns classify individual rows and can be used as grouping fields.

The date reference in Date Status should be updated when a different analysis cutoff is required. After changing it, the date classification and dependent visuals should be checked again.

16. Business Recommendations

The analysis supports several practical recommendations.

16.1 Investigate Low Regional Profitability

The Central region recorded a profit margin of 7.92%, the lowest among the four regions.

Review its product mix, discount distribution, and order-level profitability. Compare the findings with the West and East regions before deciding which operational changes may be appropriate.

16.2 Review Furniture Profitability

Furniture generated 754,747.76 in sales but only 19,730.00 in profit.

Investigate whether specific sub-categories or products contribute disproportionately to the low margin. Examine discounting and costs at a more detailed level before changing prices.

16.3 Examine High-Discount Transactions

The High discount group recorded a total loss of 136,020.69 and a profit margin of -37.29%.

Review high-discount transactions by product, sub-category, and region. Consider testing alternative discount thresholds or approval rules, while monitoring their effect on sales and profit.

16.4 Investigate Loss-Making Products

Products with repeated negative-profit transactions should be reviewed individually.

Possible actions include checking pricing, reviewing discount eligibility, validating costs, and evaluating whether the product continues to meet business objectives.

16.5 Review Future-Dated Records

The data-quality page flags 1,083 records dated after October 18, 2026.

Confirm whether the records belong to the intended reporting period. Keep valid records where appropriate and separate genuine data-quality problems from valid future-dated transactions.

16.6 Monitor Performance Through Power BI

Use the interactive dashboard to monitor KPIs by region, date, category, and sub-category. Regular monitoring can help identify changes in sales and profitability earlier than occasional static reporting.

16.7 Validate Recommendations Before Implementation

The results are descriptive. They highlight patterns and areas for investigation but do not establish the cause of every loss or regional difference.

Before implementing business changes, validate the findings using detailed transaction data, cost information, and additional analysis.

17. Limitations

Several limitations should be considered when interpreting the results.

Dataset scope: The analysis is based on the supplied Sample Superstore data. Its findings may not represent the performance of an actual retailer or another dataset.

Descriptive analysis: The project identifies associations and patterns but does not establish causal relationships between discounts, products, regions, and profitability.

Cost information: The available analysis does not independently establish the contribution of every operational cost or expense to the reported profit figures.

Date reference: Future-dated records are classified relative to October 18, 2026. The resulting count depends on this fixed reference date and on the interpretation of the dataset.

Returns: Although a Returns table is included, the dashboard's primary KPIs focus on sales and profit. A dedicated returns analysis would require additional measures and validation of the returns relationship.

Order counting: Distinct order IDs are used for the Total Orders KPI. This is different from counting individual product rows.

Selected products: The product examples in Section 10 illustrate selected loss-making records and should not be interpreted as a complete list of all loss-making products.

These limitations provide opportunities for future improvements, including more detailed cost analysis, returns analysis, validation of business keys, and deeper investigation of the drivers of profitability.

18. Conclusion

This project demonstrates how SQL, Power BI, and DAX can be combined to analyze retail sales performance and present the results in an interactive dashboard.

The dataset generated total sales of 2,326,534.35 and total profit of 292,296.82, with an overall profit margin of approximately 12.56%. Regional analysis identified the West as the leading region by sales and profit. Category-level analysis showed that Technology generated the highest total profit, while Furniture recorded a comparatively low profit margin.

Sub-category analysis highlighted negative profitability in Tables, Bookcases, and Supplies. Discount analysis revealed that the High discount group generated a substantial overall loss, making high-discount transactions an important area for further investigation.

The data-quality analysis flagged 1,083 records dated after the fixed reference date of October 18, 2026. These records should be reviewed in the context of the dataset's intended reporting period rather than automatically classified as errors.

The three-page Power BI dashboard brings these findings together in a practical reporting tool. Overall, the project demonstrates skills in data inspection, SQL aggregation, relational data modeling, DAX calculations, dashboard development, analytical interpretation, and business reporting.

19. Appendix: SQL Scripts

The following examples illustrate SQL techniques relevant to the analysis. Table and column names may need to be adjusted to match the exact names used in the MySQL database.

19.1 Total Sales and Profit

```
SELECT
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit,
    SUM(profit) / NULLIF(SUM(sales), 0) * 100 AS profit_margin_pct
FROM orders;
```

This query calculates overall sales, profit, and profit margin.

19.2 Regional Performance

```
SELECT
    region,
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit,
    SUM(profit) / NULLIF(SUM(sales), 0) * 100 AS profit_margin_pct
FROM orders
GROUP BY region
ORDER BY total_sales DESC;
```

This query summarizes sales and profitability by region.

19.3 Category Performance

```
SELECT
    category,
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit,
    SUM(profit) / NULLIF(SUM(sales), 0) * 100 AS profit_margin_pct
FROM orders
GROUP BY category
ORDER BY total_profit DESC;
```

This query compares the financial contribution of each product category.

19.4 Sub-Category Profitability

```
SELECT
    sub_category,
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit,
    SUM(profit) / NULLIF(SUM(sales), 0) * 100 AS profit_margin_pct
FROM orders
GROUP BY sub_category
ORDER BY total_profit ASC;
```

This query places the lowest-profit sub-categories first to facilitate investigation.

19.5 Classification of Discount Levels

```
SELECT
    CASE
        WHEN discount = 0 THEN 'No Discount'
        WHEN discount <= 0.10 THEN 'Low'
        WHEN discount <= 0.20 THEN 'Medium'
        ELSE 'High'
    END AS discount_level,
    COUNT(*) AS number_of_rows,
    AVG(discount) * 100 AS average_discount_pct,
    SUM(sales) AS total_sales,
```

```
    SUM(profit) AS total_profit,
    SUM(profit) / NULLIF(SUM(sales), 0) * 100 AS profit_margin_pct
FROM orders
GROUP BY
    CASE
        WHEN discount = 0 THEN 'No Discount'
        WHEN discount <= 0.10 THEN 'Low'
        WHEN discount <= 0.20 THEN 'Medium'
        ELSE 'High'
    END
ORDER BY average_discount_pct;
```

This query groups records by discount level and compares their financial results.

19.6 Identifying Loss-Making Products

```
SELECT
    product_name,
    COUNT(*) AS number_of_rows,
    AVG(discount) * 100 AS average_discount_pct,
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit
FROM orders
```

```
GROUP BY product_name

HAVING SUM(profit) < 0

ORDER BY total_profit ASC;
```

This query identifies products whose aggregated profit is negative.

19.7 Flagging Future-Dated Records

```
SELECT

    COUNT(*) AS future_dated_rows,

    SUM(sales) AS future_dated_sales,

    SUM(profit) AS future_dated_profit

FROM orders

WHERE order_date > '2026-10-18';
```

This query counts records dated after the analysis reference date and calculates their total sales and profit.

19.8 Future-Dated Records by Month

```
SELECT

    DATE_FORMAT(order_date, '%Y-%m') AS order_month,

    COUNT(*) AS number_of_rows,

    SUM(sales) AS total_sales,

    SUM(profit) AS total_profit

FROM orders

WHERE order_date > '2026-10-18'
```

```
GROUP BY DATE_FORMAT(order_date, '%Y-%m')

ORDER BY order_month;
```

This query summarizes the flagged records by month to help identify their distribution over time.

End of Report

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Project: Sales Performance Analysis — SQL and Power BI